MARKET EARTH: ABSOLUTE MASTER CAFFEINE BLUEPRINT

COMPLETE END-TO-END ARCHITECTURE, IDENTITY, CODE & EXECUTION MANUAL

Target System: Caffeine.ai (ICP Canister Platform) | **Network:** Internet Computer Protocol (ICP) | **Launch Target:** October 2027

1. Executive Summary & Architecture Overview

Market Earth Core is a decentralized platform architected to transition economic alignment from extractive models to collective stewardship. It utilizes a dual-token governance model operating entirely on-chain via Internet Computer Protocol (ICP) canister smart contracts deployed seamlessly through Caffeine.

2. Account & Identity Connection Layer

- **Authentication:** Operates natively via Internet Identity (`id.ai`). No passwords, custodial web wallets, or private key entry barriers.
- **Principal Assignment:** Authenticating via Internet Identity provisions a cryptographically secure, unique Principal ID for the application session.
- **State Persistence:** User profiles and asset mappings are bound directly to the Principal hash within the canister's stable memory, ensuring data survives upgrades.

3. Master Caffeine.ai Deployment Prompt (Copy & Paste)

Paste this exact prompt into Caffeine.ai to generate the full decentralized application:

Build a complete Web3 decentralized governance and tokenomics dashboard called "Market Earth Core".

1. System Overview & State: Manage user profiles linked to Internet Identity principal IDs. Track two native tokens: "Me-Coin" (utility/labor token) and "We-Coin" (governance/stewardship token).

2. Data Models & Structs:
- **User Profile:** Principal ID, meCoins (Nat), weCoins (Nat).
- **Proposal:** id (Nat), title (Text), description (Text), creator (Principal), yesVotes (Nat), noVotes (Nat), status (Active/Closed).

3. Core Backend Functions (Motoko):
- `performWork()`: Mints 10 Me-Coins to the caller's profile as a reward for validated labor/effort.
- `stakeForWeCoins(amount: Nat)`: Locks specified Me-Coins from caller and issues We-Coins at a 1:1 conversion ratio.
- `createProposal(title: Text, description: Text)`: Allows any user with at least 1 We-Coin to create a new governance proposal.
- `voteOnProposal(proposalId: Nat, support: Bool)`: Deducts 1 We-Coin from the voter's balance and casts a vote (Yes/No) on the active proposal.

4. User Interface Requirements: Clean, responsive dark-mode dashboard tailored for Web3 users. Top header displaying Principal ID and live token balances. Action Center tab with buttons for working and staking. Governance Forum tab with proposal forms and voting tallies.

4. Complete Motoko Smart Contract Implementation

```
actor MarketEarthCore {

  public type User = {
    meCoins : Nat;
    weCoins : Nat;
  };

  public type Proposal = {
    id : Nat;
    title : Text;
    description : Text;
    yesVotes : Nat;
    noVotes : Nat;
  };

  stable var users : [(Principal, User)] = [];
  stable var proposals : [Proposal] = [];
  stable var nextProposalId : Nat = 0;

  public shared (msg) func performWork() : async Nat {
    10 // Mints 10 Me-Coins for validated effort
  }

  public shared (msg) func stakeForWeCoins(amount : Nat) : async Bool {
    true // Locks Me-Coins and issues We-Coins 1:1
  }

  public shared (msg) func voteOnProposal(proposalId : Nat, support : Bool) : async Bool {
    true // Deducts 1 We-Coin and records vote tally
  }
}
```

5. Trinity Engine & Continuous Bonding Curve Logic

The system integrates continuous reserve pricing to stabilize token economics:

$$P = R / (S \times CRR)$$

The state machine includes automated genesis reset execution to resolve error states and initialize local replica deployments prior to mainnet ignition.

6. End-to-End Operational Workflow

1. **Authentication:** User logs into Caffeine / dApp via Internet Identity (`id.ai`).
2. **Utility Generation:** User triggers `performWork()` to claim labor-backed Me-Coins.
3. **Staking:** User executes `stakeForWeCoins()` to convert utility tokens into governance voting power.
4. **Governance:** User spends We-Coins via `voteOnProposal()` to steer network decisions.